

Module 3: Contribution Journal

ITAI 1371 - Machine Learning .

Neural Crew .

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For this lab, I first reviewed the three main types of machine learning: supervised, unsupervised, and reinforcement learning. This was important to set the stage for what kind of problem I'd be solving. I knew this lab would focus on supervised learning, so I was looking forward to diving into that.

I needed to get my environment ready. That's why I imported all the necessary libraries like pandas for data manipulation, numpy for numerical operations, matplotlib and seaborn for visualizations, and various tools from Sklearn for machine learning. Having everything at hand makes the workflow smooth.

Then, I loaded the Wine dataset. I chose this dataset because it's a classic for classification problems, and it's built right into scikit-learn, which makes it super easy to access. I immediately converted it into a pandas DataFrame because I find it much easier to work with than raw NumPy arrays, and I added columns for the wine class names to make it more readable. I checked its shape and class names to get a quick overview.

I did some initial exploratory data analysis (EDA). I looked at the overall shape, the number of features, and especially the distribution of the wine classes. It's good to see that there are no missing values, which means less cleaning for me! I also plotted the class distribution to see if the classes were balanced (they're fairly well-balanced, which is good for training) and a correlation heatmap of the first few features to understand if any features were strongly related to each other. This was the core of my lab, applying the 6-step ML workflow:

1. **Data Preparation:** For simplicity, I decided to pick just four features (alcohol, malic_acid, ash, and alcalinity_of_ash) for my initial model. I separated these features (X) from the target variable (wine_class, or y).
2. **Data Splitting:** This is a crucial step! I split my data into training (80%) and testing (20%) sets. I used random_state=42 to ensure my results were reproducible, and stratify=y to make sure the class proportions were maintained in both the training and testing sets. This prevents the model from just memorizing the training data and gives me a way to evaluate its performance on unseen data.
3. **Model Training:** I chose to train two different classification models: Logistic Regression and a Decision Tree. Logistic Regression is a good baseline linear model, and a Decision Tree is a non-linear model. I set random_state=42 for both to

ensure reproducibility and limited the max_depth of the Decision Tree to 3 to prevent it from becoming too complex and overfitting early on.

4. **Model Evaluation:** After training, I needed to see how well my models performed. I used the held-out test set to make predictions and then calculated the accuracy score for both models. I also generated a detailed classification_report which gives me precision, recall, and F1-score for each class, which is much more informative than just accuracy, especially if classes were imbalanced. Based on this, Logistic Regression turned out to be the better performer with my chosen features.
5. **Model Interpretation:** To understand how the best model was making its predictions and where it was making mistakes, I visualized the confusion matrix for the Logistic Regression model. The diagonal tells me correct predictions, and the off-diagonal cells show misclassifications. It's a great way to see which classes are being confused with each other.

I also took a moment to review different data types like continuous, discrete, nominal, ordinal, text, and boolean. This section reminded me that the type of data dictates how you preprocess it and what algorithms are most suitable.

Hands-On Practice

For hands-on practice, I tried selecting a different set of features (alcohol, color_intensity, proline) to see if I could improve the model's accuracy. It's an iterative process, and trying different features is a fundamental part of optimizing performance. My new combination didn't beat the original, but it showed me the impact of feature choice!

I believe that I succeeded by identifying ML types, which was a good check of my foundational knowledge. Finally, the reflection section The most important thing I learned throughout this lab was the importance of the entire ML workflow, from data understanding to iterative evaluation.

The most challenging part was truly understanding evaluation metrics beyond just accuracy, like the nuances of the confusion matrix. Moving forward, in the future, I would love to explore how to handle imbalanced datasets, systematic feature selection, and effective hyperparameter tuning without overfitting.